

The vowel that shifted most dramatically in connected speech was /æ/, showing a marked centralizing drift in F_1 and F_2 compared to its isolated /hVd/ baseline. In continuous speech, shorter duration prevented the tongue and jaw from reaching extreme peripheral positions, leading to articulatory target undershoot. Consequently, F_1 decreased due to reduced jaw opening, while F_2 shifted toward a mid-range frequency typical of schwa-like central reduction. In contrast, vowels in isolated /hVd/ contexts retained higher acoustic clarity and distinct formant targets due to hyperarticulated citation forms.